

Anti-epileptic activity of *Nelumbo nucifera* fruit

Muhammad Ali Rajput¹ · Rafeeq Alam Khan¹ · Tahira Assad¹

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Abstract Epilepsy is the most commonly encountered neurological disorder affecting around 70 million people worldwide, out of which approximately 80% belongs to developing countries. Several shortcomings appeared with the use of conventional antiepileptic agents like, inadequate seizure control, side effects and cost which limit their use. Thus extensive studies are necessary to investigate the pharmacological effects of plants, which would facilitate discovery of novel drugs from herbal source permitting their use to benefit mankind. Hence current study was focused to evaluate the anti-epileptic potential of *Nelumbo nucifera* fruit in order to ascertain its therapeutic potential. Anti-epileptic activity was assessed using strychnine induced seizure model in 35 male Wistar rats divided in five groups i.e. control, reference and 3 test groups. Each group was composed of 7 animals and was given 2% gum tragacanth (control), diazepam 1 mg/kg PO (reference) and *N. nucifera* fruit 50, 100 and 200 mg/kg PO (test) OD for 15 days. *N. nucifera* fruit extract at 200 mg/kg exhibited extremely noteworthy delay in the inception of convulsions as compared to control however duration of convulsions was increased significantly but intensity of convulsions was reduced resulting in better survival rate i.e. 42.85% which was comparable to diazepam. Therefore it can be concluded that *N. nucifera* fruit may be valuable in managing epilepsy but further studies are required on large number of animals to confirm these findings.

Keywords Strychnine · Gum tragacanth · Diazepam · *Nelumbo nucifera*

Introduction

Epilepsy is a mental or neurologic disorder that encircles wide range of conditions which results in dysfunction of brain and spinal cord (Dennis et al. 2003). It is one of the significant medical complications of today affecting approximately thirty thousand people around the globe each year. This condition hits around 1 in 20 individuals in their lifetime adding about 20–70 fresh cases of epilepsy every year in 100,000 individuals (Shporvan 1990).

Evidence proposed that inequity between excitatory and inhibitory synaptic transmission within CNS is a major cause of epilepsy in both experimental and clinical situations (Perez-Saad and Buzneqo 2008). There are various classes of anti-epileptic's that have clinical effectiveness and have shown fine prognosis in controlling convulsions in majority of patients (Cockerell et al. 1995). Regardless of that lot of patients with epilepsy were never managed with conventional agents. Furthermore, the high rate of occurrence of undesirable effects, contraindications and possible interactions with the use of established antiepileptic drugs (AEDs) is also a matter of distress for individuals who utilize them consistently (Perucca et al. 1993). That's why more people are switching towards safer alternatives, specially derived from plants (Rajput et al. 2012).

N. nucifera belongs to family *Nymphaeaceae* is widely cultivated in the tropical regions of Pakistan, India, China, Thailand and Australia (Mukherjee et al. 2009). *N. nucifera* fruit consists of seeds and seed pods. The latter is also known as lotus bulbs, are green in color and provides attachment to the seeds, which are black in color, hard in nature, ovoid,

✉ Rafeeq Alam Khan
rkhan1959@gmail.com

¹ Department of Pharmacology, Faculty of Pharmacy and Pharmaceutical Sciences, University of Karachi, Karachi 75270, Pakistan

oblongish or roundish in shape, up to 1.5 cm broad and 1.0 cm long and are organized in whorls (Sridhar and Bhat 2007). The seeds are the edible portion and are studded in the pod or bulb and have to be peeled individually before they are eaten. The seed is perhaps the strongest source of protein in the plant (Moro et al. 2013).

The *N. nucifera* seeds are loaded with fat, amino acid, protein, asparagines, unsaturated fatty acids, minerals, starch, saponins and tannins. The seed has three main components; cotyledons, plumule and integuments, which occupies approximately 93.23, 3.03 & 3.74% in terms of its mass correspondingly. The average burden of hundred seeds is approximately 87.35 g. The seeds are rich in calcium, potassium and magnesium, and contain various essential bio metals such as sodium, iron, chromium, manganese, zinc and copper. The seed are highly nutritious and possesses 348.45 cal per 100 g (Indrayan et al. 2005).

The qualitative phytochemical analysis of the seed pods of *N. nucifera* fruit exposed the presence of flavonoids, alkaloids, saponins, tannins and terpenoids (Rajput and Khan 2017). Procyanidin was also identified and sequestered from the pods of *N. nucifera* fruit. The chief secondary metabolites present in *N. nucifera* seeds are alkaloids, particularly lotusine, liensinine, isoliensinine, dauricine, nuciferine, pronuciferine, roemerine, procyanidin, neferine and arnepavine. The seeds also have carbohydrates, Gallic acid and isoquininolinol (Mukherjee et al. 2009).

Traditionally fruits of *N. nucifera* are consumed as a healthy foodstuff in Asia and for the cure of numerous conditions such as fever, inflammation, insomnia, palpitation, hypertension, arrhythmia, enteritis, chronic diarrhea, spermatorrhea, leucorrhoea, halitosis, skin diseases, leprosy and menorrhagia. It is also used as an antiemetic, antidote, refrigerant and diuretic (Chopra et al. 1956; Vershney and Rzoska 1976). Recent studies demonstrate its anxiolytic and antidepressant effect (Rajput and Khan 2017) but very limited literature is available regarding its anti-epileptic effect, thus current study was focused to ascertain the anti-epileptic potential of *N. nucifera* fruit. This study would surely help in the exploration of a novel agent from the plant source and therefore will recommend the applications of complementary medicines in humans (Ahmed et al. 2015).

Materials and methods

Experimental design & methodology

Research work was executed utilizing the laboratory facilities of Pharmacology department and the Research Institute of Pharmaceutical Sciences, University of Karachi, following approval from the Board of Advance Studies & Research (BASR), and departmental research committee.

Selection of animals

35 male Wistar rats were used for the evaluation of antiepileptic activity and were divided into five treatment groups i.e. control, reference and 3 test groups. 7 rats were kept in each group.

Animal housing

Animals were placed in plastic cages with preservation of room temperature at 23 ± 2 °C and moisture kept at 50 to 60% in a substituting 12-h light/dark succession. Every animal was provided with normal diet prepared in laboratory plus water as desired. The animals were shifted to the laboratory approximately an hour before the start of experiments. The experiment was performed during day time. Prior to dose administration, overall fitness of animals were assessed throughout the adaptation phase utilizing the laboratory conditions for a week particularly observing lack of activity, edema, diarrhea and ulceration.

The research committee Department of Pharmacology permitted the use of animals for these experiments in agreement with the guidelines of NIH (The National Institute of Health Guide for the Care and Use of Laboratory Animals 2010) and NACLAR (National Advisory Committee for Laboratory Animal Research 2004).

Plant material and preparation of extract

N. nucifera fruits were obtained from the local fruit & vegetable market of Qasimabad, Hyderabad, Sindh, Pakistan. The *N. nucifera* fruits were identified and authenticated in the Department of Pharmacognosy. The receipt sample no NNF-03 was placed in the section of Pharmacognosy, University of Karachi.

Grounding of crude extract followed cold extraction procedure (Hossain et al. 2010). 6 kg fruits were washed and the seeds were peeled off from the seed pods of the fruits manually and were then chopped. After that chopped seeds were dried under shade for 6 days. The dried material was ground to coarse powder. The seed pods on the other hand were dried under shade for 3 days and then chopped to fine powder. The seed powder along with the seed pod powder was macerated in 10 l of 98% ethanol for 30 days with occasional shaking and stirring until the color of the solvent becomes black.

The solvent was filtered with the help of Whatman No. 1 filter paper. Afterwards the extract was evaporated under reduced pressure in a rotary evaporator at 40 °C to 45 °C followed by freeze drying at −30 °C. The solid lyophilized material so obtained was saved at −20 °C until further use in doses of 50, 100 and 200 mg/kg orally (Rajput and Khan 2017). The final quantity of the extract obtained was 400 g of dry mass.

Preparation of drugs

The dilution of strychnine was made in sterile water and then given intraperitoneally (IP) in a dose of 4 mg/kg to provoke seizures (Duraismi et al. 2009).

The suspending agent (*gum tragacanth*) was consumed to prepare suspensions of the control, reference drug (diazepam) and 3 doses of test group (*N. nucifera* fruit extract 50, 100 & 200 mg/kg). Gum tragacanth was administered to control group as placebo in the dose of 10 ml/kg by oral route. Every time fresh suspensions were made for the purpose of dosing (Madhu et al. 2009; Rajput et al. 2013).

Diazepam 5 mg tablets were trampled to powder and then suspended in 2% gum tragacanth, administered orally in a dose of 1 mg/kg (Martinez et al. 2006) with the help of orogastric tube.

Strychnine induced seizures

35 male rats were divided in five groups i.e. control, reference and 3 test groups. Each comprising of 7 animals and were given 2% gum tragacanth (control), diazepam 1 mg/kg PO (reference) and *N. nucifera* fruit 50, 100 and 200 mg/kg PO (test) OD for 15 days. On 15th day of study, strychnine in a dose of 4 mg/kg IP was administered to rats 40 min after the administration of gum tragacanth, diazepam and fruit extract at specified doses and time of onset of seizures along with period the of convulsions and the percentage of animals survived after strychnine administration was recorded.

Strychnine exerts its convulsant action by blocking directly the inhibitory CNS (spinal cord and brain stem) reflexes of glycine and thus enhances spinal nerve reflexes (Biggio et al. 1992). Rats that escape convulsions thirty minutes after strychnine administration were considered protected (Duraismi et al. 2009).

Statistical analysis

The data was subjected to analysis by taking average and standard error to the average utilizing two sample student T-test and values of $P < 0.05$ were considered as noteworthy and $P < 0.005$ as extremely noteworthy. All statistical techniques were executed using SPSS software version 20 (Walpole 1982).

Results

Strychnine induced seizures

Table 1 has depicted the effect of *N. nucifera* fruit and diazepam on epilepsy using strychnine induced seizure model in rats and time of onset of convulsions, duration of convulsions

and percentage of animals remain protected or avoided seizures were recorded. *N. nucifera* fruit ethanol extract at dose 200 mg/kg exhibited extremely noteworthy delay in the inception of convulsions in comparison with control while duration of convulsions was also increased significantly at the same dose of test drug as compared to control but the intensity of convulsions was reduced which resulted in better percentage of animal survived i.e. 42.85%. *N. nucifera* fruit extract at 100 mg/kg also showed noteworthy delay in the onset of convulsions as compared to control without exhibiting any significant change in the duration of convulsions in comparison to control however resulted in the survival of 14.28% animals. The time for the onset of convulsions was also deferred and period of convulsions was also increased with 50 mg/kg *N. nucifera* fruit extract but less than higher doses. However diazepam 1 mg/kg considerably delayed the onset of convulsions; while duration of convulsions was also significantly increased as compared to control and the survival rate was 57.14%.

Discussion

Epilepsy is the second most commonly encountered neurological disorder affecting around 70 million people worldwide, out of which approximately 80% are in developing countries (Yemedje et al. 2011). Several short comings appeared with the use of conventional AEDs like, inadequate seizure control, side effects, cost and potentiation of epilepsy induced comorbidities such as depression, memory deficits and mood disorders, which limit their use as comprehensive therapy. Therefore, satisfactory treatment of epilepsy not only demands the suppression of abnormal neuronal discharge but also control of epileptic co-morbidities associated with it along with wide margin of safety and low cost (Paramdeep et al. 2014).

Present study revealed the antiepileptic effect of *N. nucifera* fruit and diazepam using strychnine induced seizure model in rats. There was delay in onset of seizures, increase in duration of convulsions and increase in percentage of animals survived from convulsions. *N. nucifera* fruit extract exhibited pronounced antiepileptic activity at a dose of 200 mg/kg which resulted in highly significant delay in the onset of convulsions and increased the survival of animals from seizures i.e. 42.85%.

In the central nervous system various flavonoids have been found to inhibit almost all the mechanisms involved in seizures generation in epilepsy e.g. inhibition of voltage gated Na^+ channels results in decrease sodium ion influx into the neuronal cell (Nicholson et al. 2010); activation of Ca^{2+} activated K^+ channels results in increased potassium ion outflow from the neuronal cell (Engelborghs et al. 2000); activation of inhibitory GABAergic receptors through direct action on

Table 1 Effect of *N. nucifera* fruit and diazepam on strychnine induced convulsions in rats

Groups/Doses	Onset of convulsions (minutes)	Duration of convulsions (minutes)	Animals survived or protected (%)
Control	0.1 ± 0.01	0.9 ± 0.41	0
10 ml/kg <i>N. nucifera</i> fruit	0.4 ± 0.15	1.3 ± 0.35	14.28
50 mg/kg <i>N. nucifera</i> fruit	1.8 ± 0.42*	1.6 ± 0.71	14.28
100 mg/kg <i>N. nucifera</i> fruit	2.8 ± 0.27**	1.6 ± 0.22*	42.85
200 mg/kg Diazepam	2.8 ± 0.69*	2.4 ± 0.45*	57.14
1 mg/kg			

$n = 7$

Values are Mean ± S.E.M

* $p < 0.05$ noteworthy in comparison to control

** $p < 0.005$ extremely noteworthy in comparison to control

GABA or through benzodiazepine receptor results in increased chloride ion influx causing neuronal hyperpolarisation (Hanrahan et al. 2011); inhibition of opioid receptors leading to proconvulsant or anticonvulsant effects (Engelborghs et al. 2000); Inhibition of NMDA receptor, results in decrease entry of calcium into the neuronal cell (Subash and Subramanian 2009); and antioxidant effects of flavonoids leading to increased seizure threshold (Paramdeep et al. 2014).

Progression of epilepsy is also dependent on the development of free radicals. Raised levels of free radicals are coupled with low level of antioxidants, which results in the neuronal damage. It is already documented that development of free radicals provoke convulsions through inhibiting enzyme glutamine synthase which leads to accumulation of stimulatory neurotransmitter, glutamate or by decreasing glutamate decarboxylase, thus causing decrease in GABA turnover (Devi et al. 2008). Considering this, flavonoids are a bioactive metabolite of choice for the treatment of oxidative stress induced diseases like epilepsy. Flavonoids, due to their antiepileptic and neuroprotective effects have been found to ameliorate the co-morbidities. They interact with critical protein and lipid kinase signaling cascades e.g. phosphatidylinositol-3 kinase (PI3K)/Akt, protein kinase C and mitogen-activated protein kinase in the brain, leading to inhibition of apoptosis activated by neurotoxic species and promote neuronal endurance and synaptic plasticity. Moreover, flavonoids exert positive impact on the vascular system which results in modification in cerebral blood supply and are able enough to speed up angiogenesis, neurogenesis and transformations in neuronal morphology (Vauzour et al. 2008).

Procyanidin and tannins which are also present in the seed pod of *N. nucifera* fruit have demonstrated antioxidant effects in previous studies which may also contribute in reducing the oxidative stress by lowering the levels of free radicals in the brain responsible for neuronal damage (Ling et al. 2005).

Similarly terpenoids in various studies have exhibited alteration in GABA mediated system and diminution in neuronal excitability by jamming the voltage dependent sodium channels (De Almeida et al. 2008). Saponins loaded fractions of *ficus platyphyla* stem bark have also exhibited anticonvulsant activity on strychnine induced and PTZ seizure models (Chindo et al. 2009). Moreover recent study on *N. nucifera* fruit has demonstrated its anti-depressant effect which was postulated to be due to the presence of alkaloids such as neferine, liensinine and isoliensinine in the seeds of the fruit (Rajput and Khan 2017).

N. nucifera fruit is a wealthy source of flavonoids, alkaloids, saponins tannins and terpenoids which have shown anti-epileptic activity in various studies (Kumar and Sharma 2005; Silvano et al. 2016; Zhu et al. 2014), so it may be suggested that *N. nucifera* fruit exerts its antiepileptic effect through synergistic action of all of these constituents.

Conclusion

N. nucifera fruit have exhibited strong antiepileptic effect which may be due to rich flavonoids contents of nucifera fruit, since flavonoids had shown to have anticonvulsant effects. Therefore it can be concluded that *N. nucifera* fruit may be beneficial in managing epilepsy and the co-morbidities associated with it such as anxiety and depression but further studies are required on large number of animals to confirm these findings.

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